

Video-Based Pediatric Seizure Detection from Pose Landmark Sequences

An ensemble of CNN – GRU and Temporal Convolutional Networks with engineered kinematic features · binary detection of infantile spasms (West syndrome) from anonymised MediaPipe Pose

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Task — binary detection of infantile spasms from 5-second pose-only segments ($150 \times 33 \times 5$ MediaPipe tensors). **Approach** — 5-fold ensemble of CNN-GRU + multi-scale TCN over 518-d engineered features, focal loss, child-grouped cross-validation, 7-variant TTA. **Result** — mean CV F_1 0.628 / 0.635; held-out test F_1 **0.2975** (rank 19 of 26; top score 0.4502). **Code** — github.com/babowoyele/seizure-detection

1. Problem

Infantile spasms are the defining seizure type of West syndrome, an age-dependent epileptic encephalopathy with peak onset around six months of life. A spasm is a brief (1–3 s) sudden contraction of trunk and limbs — flexor, extensor, or mixed — typically clustered near sleep–wake transitions and frequently first observed by caregivers in home-video.

Two constraints shape a deployable system:

- Privacy.** Pixel-level video carries identity, room context, and clinical metadata.
- Long recordings.** Home and clinical recordings span hours to days; raw-pixel pipelines do not scale.

A *pose-only* intermediate — a few hundred floats per frame, invariant to lighting, clothing, and background — solves both. The 2026 challenge distributes exactly this representation.

2. Data

Per-segment files: **(150 frames, 33 landmarks, 5 channels)** — anonymised MediaPipe Pose. Channels per landmark: x, y, z , *visibility*, *presence*. Filenames carry a child-ID we exploit for leakage-free grouped cross-validation.

- ≈5 s at 30 fps per segment.
- Strong class imbalance toward non-seizure.
- Frame-level missingness handled with forward/back-fill and an explicit binary mask feature.

3. Preprocessing

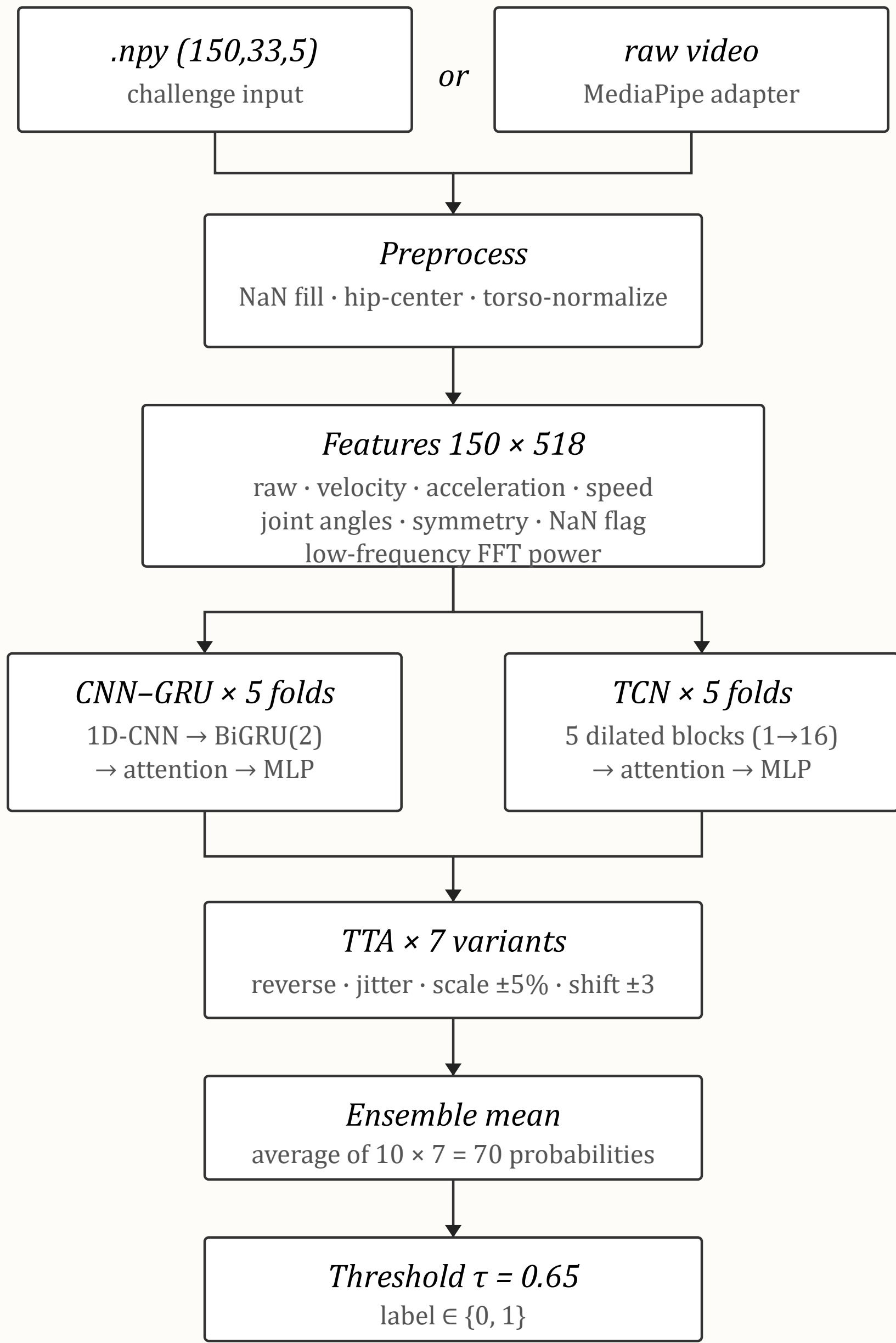
- NaN handling.** Forward- then back-fill missing frames; preserve the binary NaN mask as a feature.
- Hip-centering.** Translate so the mid-hip point (landmarks 23, 24) sits at the origin per frame — translation-invariant.
- Torso normalisation.** Divide x,y,z by the per-frame mid-shoulder→mid-hip distance — scale-invariant against camera-distance.

4. Features (518 per frame)

Component	Dim.
Raw landmarks ($x,y,z,\text{vis},\text{pres}$)	165
Velocity (Δxyz)	99
Acceleration ($\Delta^2 xyz$)	99
Speed magnitude per landmark	33
Joint angles (10 triplets)	10
Left–right symmetry (6 pairs)	6
NaN-frame indicator	1
FFT power (7 lm $\times$ 3 axes $\times$ 5 bins)	105
Total	518

Spectral bins band-limited to ≈0.2–3 Hz — the rhythmic range of clonic/myoclonic spasm activity at 30 fps over a 5-s window.

5. Pipeline



End-to-end pipeline. Either pre-extracted .npy arrays or raw video enter a shared preprocessing + feature stage; two architectures are ensembled across 5 folds and 7 TTA variants.

6. Models

CNN-GRU. Two 1-D conv layers ($k=5, k=3$) project 518-d $\rightarrow$ 128 channels, then a 2-layer bidirectional GRU (hidden=128), then learned attention pooling, then a $256 \rightarrow 64 \rightarrow 1$ MLP head.

Multi-scale TCN. Five residual blocks with kernel 3 and dilations {1, 2, 4, 8, 16} — full receptive-field coverage of the 150-frame window — then attention pooling and the same MLP head.

Why two. Recurrent vs. parallel multi-scale inductive biases produce de-correlated errors and different optimal thresholds (see Results).

7. Training

- 5-fold StratifiedGroupKFold by child-ID** — no within-subject leakage.
- Focal loss** ($\alpha=0.75, \gamma=2.0$) over a class-balanced WeightedRandomSampler.
- AdamW** + OneCycleLR ($\text{max_lr} = 5 \times 10^{-3}$, $\text{pct_start} = 0.3$).
- Up to 120 epochs, early stopping on val F_1 (patience 25).
- Augmentation: time-reverse, jitter, time-resample, frame dropout, same-class mixup.

8. Cross-validation results

Fold	CNN-GRU F_1	$\hat{t}$	TCN F_1	$\hat{t}$
0	0.633	0.70	0.611	0.63
1	0.615	0.59	0.608	0.61
2	0.629	0.64	0.635	0.33
3	0.575	0.28	0.627	0.25
4	0.691	0.51	0.694	0.39
Mean	0.628	0.54	0.635	0.44

Per-fold optimal thresholds for CNN-GRU cluster high (0.51–0.70); TCN often low (0.25–0.39). Different operating points on the precision–recall curve — exactly the model diversity an ensemble exploits.

9. Held-out leaderboard

Rank	Team	F_1	Approach
1	Kramer (BU/JHU)	0.4502	Two-step likelihood + classifier
2	Ramesh (NIT Tiruchirappalli)	0.4286	1D-CNN + LightGBM
3	Daraie (JHU)	0.4048	1D-ResNet + XGBoost
4	Walsh (Penn)	0.3984	GCN + bidirectional GRU
5	Hazra (IIT Kharagpur)	0.3923	ST-GCN + attention-MIL ensemble
⋮	⋮	⋮	⋮
19	Owoyele et al. (HPI, this work)	0.2975	CNN-GRU + TCN ensemble

26 valid submissions; no entry exceeded F_1 0.5. The leaderboard is compressed: rank 1 to rank 10 spans <12 percentage points.

10. Discussion — the CV→test gap

Cross-validation $F_1 \approx 0.63 \rightarrow$ held-out 0.2975 is a substantial gap. Five plausible contributors:

- Distribution shift** between training-pool and test-pool patients; grouped CV controls leakage but not pool shift.
- Threshold over-fit** to CV folds — per-fold $\hat{t}$ spans [0.25, 0.70].
- F_1 brittleness** under strong imbalance.
- Train/test TTA mismatch:** aggressive train augmentation, near-identity TTA.
- Architectural mismatch:** full-window attention may under-exploit the sub-second concentration of spasm motor energy that explicit MIL pooling captures.

11. Future work

- MeTRAbs** as pose front-end: metric-scale, 87-joint 3-D coordinates exposing hand and head dynamics.
- Multi-view triangulation** via markerless dynamic extrinsic calibration [1] — fuse synchronised views, remove occlusion failure modes.
- MIL-style sub-window pooling** matched to 1–3 s spasm duration.
- Held-out fold for threshold calibration** (rather than CV-direct).
- Integration with MaskAnyone** [2] — caregiver home-video processed entirely on-device; only landmarks and labels leave the premises.